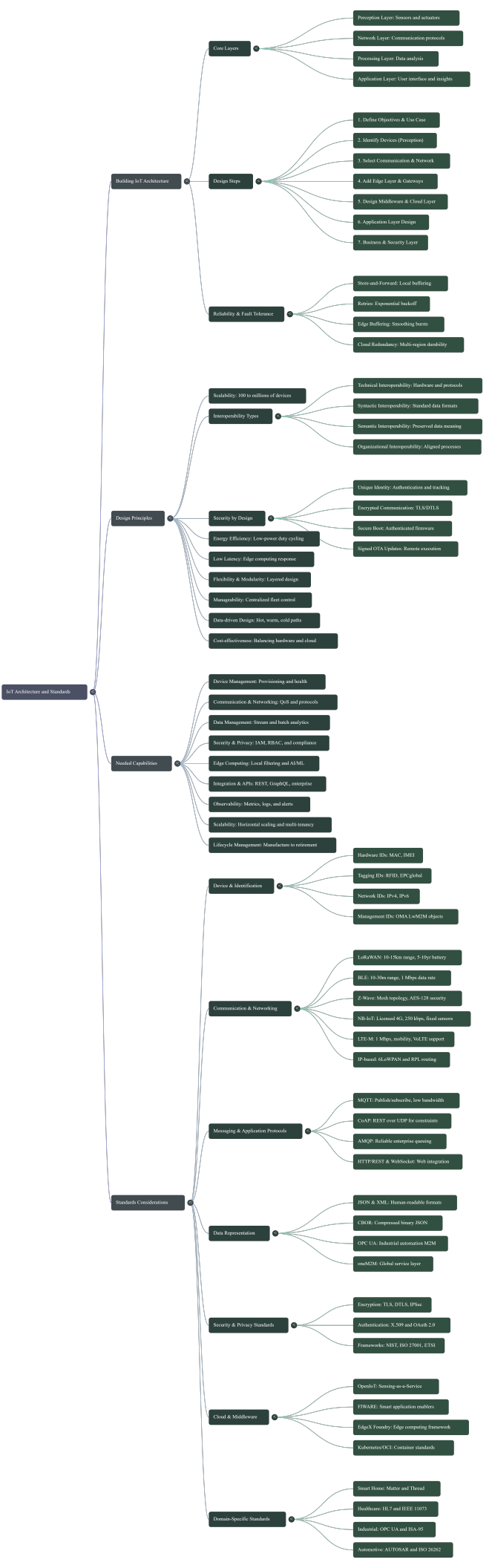

IoT Architecture and Standards
Building IoT Architecture
Core Layers
Perception Layer: Sensors and actuators
Network Layer: Communication protocols
Processing Layer: Data analysis
Application Layer: User interface and insights
Design Steps
1. Define Objectives & Use Case
2. Identify Devices (Perception)
3. Select Communication & Network
4. Add Edge Layer & Gateways
5. Design Middleware & Cloud Layer
6. Application Layer Design
7. Business & Security Layer
Reliability & Fault Tolerance
Store-and-Forward: Local buffering
Retries: Exponential backoff
Edge Buffering: Smoothing bursts
Cloud Redundancy: Multi-region durability
Design Principles
Scalability: 100 to millions of devices
Interoperability Types
Technical Interoperability: Hardware and protocols
Syntactic Interoperability: Standard data formats
Semantic Interoperability: Preserved data meaning
Organizational Interoperability: Aligned processes
Security by Design
Unique Identity: Authentication and tracking
Encrypted Communication: TLS/DTLS
Secure Boot: Authenticated firmware
Signed OTA Updates: Remote execution
Energy Efficiency: Low-power duty cycling
Low Latency: Edge computing response
Flexibility & Modularity: Layered design
Manageability: Centralized fleet control
Data-driven Design: Hot, warm, cold paths
Cost-effectiveness: Balancing hardware and cloud
Needed Capabilities
Device Management: Provisioning and health
Communication & Networking: QoS and protocols
Data Management: Stream and batch analytics
Security & Privacy: IAM, RBAC, and compliance
Edge Computing: Local filtering and AI/ML
Integration & APIs: REST, GraphQL, enterprise
Observability: Metrics, logs, and alerts
Scalability: Horizontal scaling and multi-tenancy
Lifecycle Management: Manufacture to retirement
Standards Considerations
Device & Identification
Hardware IDs: MAC, IMEI
Tagging IDs: RFID, EPCglobal
Network IDs: IPv4, IPv6
Management IDs: OMA LwM2M objects
Communication & Networking
LoRaWAN: 10-15km range, 5-10yr battery
BLE: 10-30m range, 1 Mbps data rate
Z-Wave: Mesh topology, AES-128 security
NB-IoT: Licensed 4G, 250 kbps, fixed sensors
LTE-M: 1 Mbps, mobility, VoLTE support
IP-based: 6LoWPAN and RPL routing
Messaging & Application Protocols
MQTT: Publish/subscribe, low bandwidth
CoAP: REST over UDP for constraints
AMQP: Reliable enterprise queuing
HTTP/REST & WebSocket: Web integration
Data Representation
JSON & XML: Human-readable formats
CBOR: Compressed binary JSON
OPC UA: Industrial automation M2M
oneM2M: Global service layer
Security & Privacy Standards
Encryption: TLS, DTLS, IPSec
Authentication: X.509 and OAuth 2.0
Frameworks: NIST, ISO 27001, ETSI
Cloud & Middleware
OpenIoT: Sensing-as-a-Service
FIWARE: Smart application enablers
EdgeX Foundry: Edge computing framework
Kubernetes/OCI: Container standards
Domain-Specific Standards
Smart Home: Matter and Thread
Healthcare: HL7 and IEEE 11073
Industrial: OPC UA and ISA-95
Automotive: AUTOSAR and ISO 26262